

TEMPERATURE-TOLERANCE CHECKING SYSTEM

■ ANIRBAN SENGUPTA AND
PALLABI SARKAR

Most of instruments and machines used in industries and research laboratories have a temperature-tolerance limit. These can neither be subjected to too high nor too low a temperature for their proper functioning. But they need to function accurately as they are always used in critical applications.

Here we describe a temperature-tolerance checking system that detects whether the temperature of an instrument is exceeding the allowable temperature limit. As soon as the temperature of the instrument exceeds the maximum permissible limit, it automatically generates an audio-visual warning.

The system displays the temperature of the instrument using ten LEDs, with each LED representing 1°C step forming a bargraph display. The last LED (LED10) indicates the highest temperature attained. Two buzzers have been used in the circuit:

One sounds when 50 per cent of tolerance temperature is reached. The other generates a warning call when LED10 glows, indicating that the maximum temperature limit has reached and the instrument has to be turned off.

Temperature sensor LM35 (IC1) used in the circuit is calibrated directly in degrees Celsius and its output is linearly proportional to the temperature it senses—usually 10 mV/°C. It is rated for the full range of -55°C to +150°C. The LM3914 (IC2) is a monolithic integrated circuit that senses the analogue voltage levels to drive the ten LEDs, providing a linear analogue display. As shown in the figure, the output of temperature sensor IC1 is fed to pin 5 of LED-driver IC2. The LED driver has ten internal comparators, whose outputs are connected to LED1 through LED10.

Two NE555 ICs have been used: The first NE555 (IC3) is configured as a monostable multivibrator. When the instrument's temperature crosses the maximum tolerance limit, LED10

glows, piezobuzzer PZ2 sounds and pin 2 of IC3 goes low to trigger it. As a result, output pin 3 of IC3 goes high to enable the astable multivibrator wired around the second NE555 (IC4). Once triggered, IC4 generates a frequency at its output pin 3 that drives transistor T1. Transistor T1, in turn, drives an 8-ohm loudspeaker (LS1) to produce audible sound. The frequency of the sound can be varied by changing the values of resistors R7 and R8 and capacitor C5.

Assemble the circuit on a general-purpose PCB and house in a suitable cabinet. Connect the sensor through wire and place it close to the instrument whose temperature tolerance is to be checked. Use a 9V battery to power the circuit. Alternatively, a standard 9V, 250mA adaptor can also be used.

Calibrating the temperature-tolerance checking system is very important. Should you want to use it for an instrument having tolerance of 10°C and normal working temperature of 55°C. Calibrate the

